

Module-05: Assignment

Q1. JavaScript Coding Practice

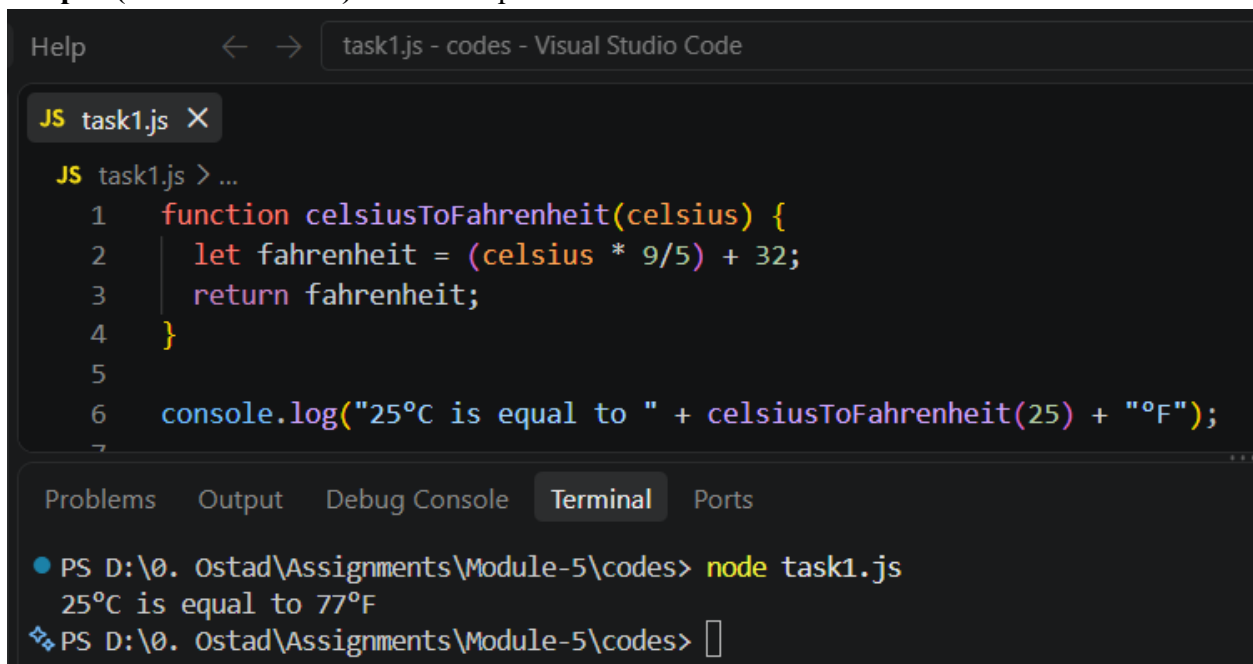
1. Problem statement:

Convert Celsius to Fahrenheit.

Source Code:

```
function celsiusToFahrenheit(celsius) {  
  let fahrenheit = (celsius * 9/5) + 32;  
  return fahrenheit;  
}  
  
console.log("25°C is equal to " + celsiusToFahrenheit(25) + "°F");
```

Output (with Screenshot): 25°C is equal to 77°F



The screenshot shows the Visual Studio Code interface. The top bar indicates the file is 'task1.js' in the 'codes' folder. The editor window shows the JavaScript code for converting Celsius to Fahrenheit. Below the editor, the 'Terminal' tab is active, showing the command 'node task1.js' being executed in a PowerShell prompt. The output of the command is '25°C is equal to 77°F'.

```
Help  <  >  task1.js - codes - Visual Studio Code  
  
JS task1.js X  
JS task1.js > ...  
1  function celsiusToFahrenheit(celsius) {  
2    let fahrenheit = (celsius * 9/5) + 32;  
3    return fahrenheit;  
4  }  
5  
6  console.log("25°C is equal to " + celsiusToFahrenheit(25) + "°F");  
7  
Problems  Output  Debug Console  Terminal  Ports  
● PS D:\0. Ostad\Assignments\Module-5\codes> node task1.js  
25°C is equal to 77°F  
❖ PS D:\0. Ostad\Assignments\Module-5\codes> 
```

Explanation:

The temperature in Celsius is taken as input, and the formula $(C \times 9/5) + 32$ is applied to it. The result is stored in a variable called fahrenheit, which is then returned and printed using console.log

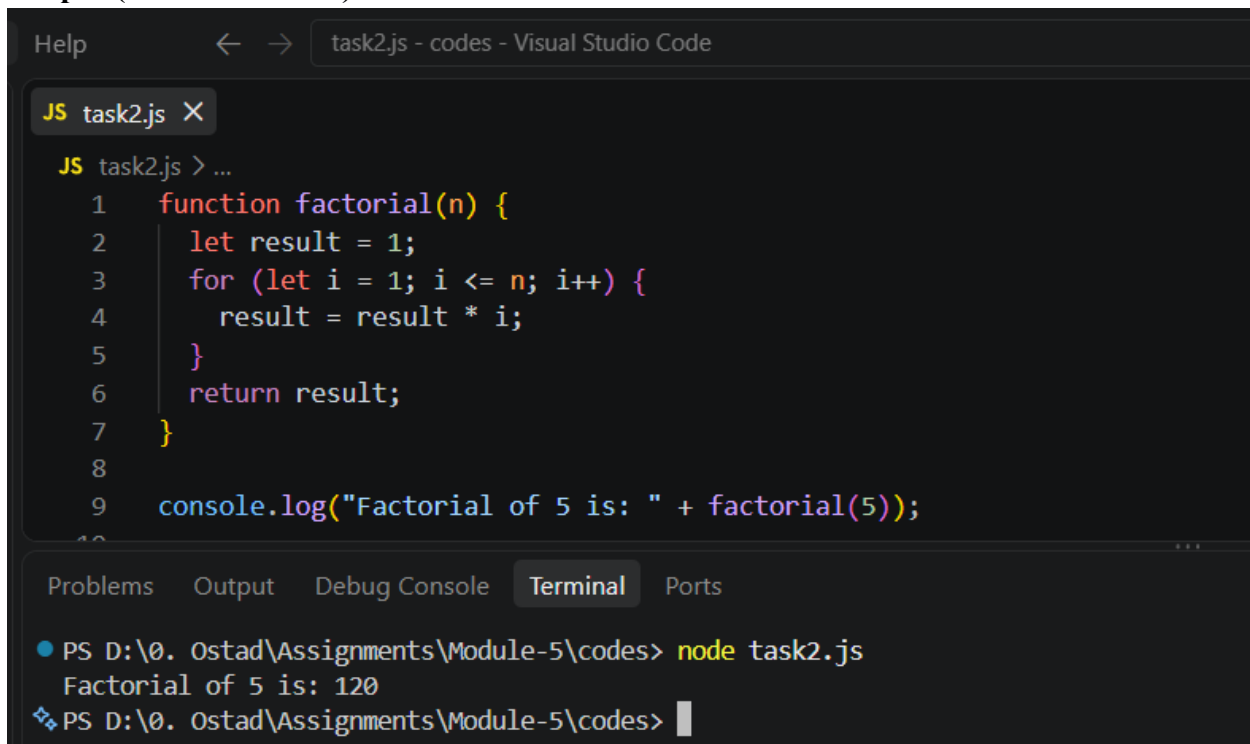
2. Problem Statement:

Find the Factorial of a Number.

Source Code:

```
function factorial(n) {  
  let result = 1;  
  for (let i = 1; i <= n; i++) {  
    result = result * i;  
  }  
  return result;  
}  
  
console.log("Factorial of 5 is: " + factorial(5));
```

Output (with Screenshot): Factorial of 5 is: 120



The screenshot shows the Visual Studio Code editor with a file named 'task2.js'. The code in the editor is a JavaScript function to calculate the factorial of a number 'n'. It initializes a 'result' variable to 1 and uses a 'for' loop to multiply 'result' by each integer from 1 to 'n'. Finally, it logs the result to the console. The terminal at the bottom shows the command 'node task2.js' being executed, resulting in the output 'Factorial of 5 is: 120'.

```
Help  <  >  task2.js - codes - Visual Studio Code  
  
JS task2.js X  
JS task2.js > ...  
1  function factorial(n) {  
2    let result = 1;  
3    for (let i = 1; i <= n; i++) {  
4      result = result * i;  
5    }  
6    return result;  
7  }  
8  
9  console.log("Factorial of 5 is: " + factorial(5));  
10  
  
Problems  Output  Debug Console  Terminal  Ports  
● PS D:\0. Ostad\Assignments\Module-5\codes> node task2.js  
Factorial of 5 is: 120  
PS D:\0. Ostad\Assignments\Module-5\codes> 
```

Explanation:

The variable `result` is initialized to 1, since multiplying by 1 doesn't change anything. A `for` loop is run from 1 to `n`, and `result` is multiplied by each number along the way. For 5, the value $1 \times 2 \times 3 \times 4 \times 5 = 120$ is computed this way.

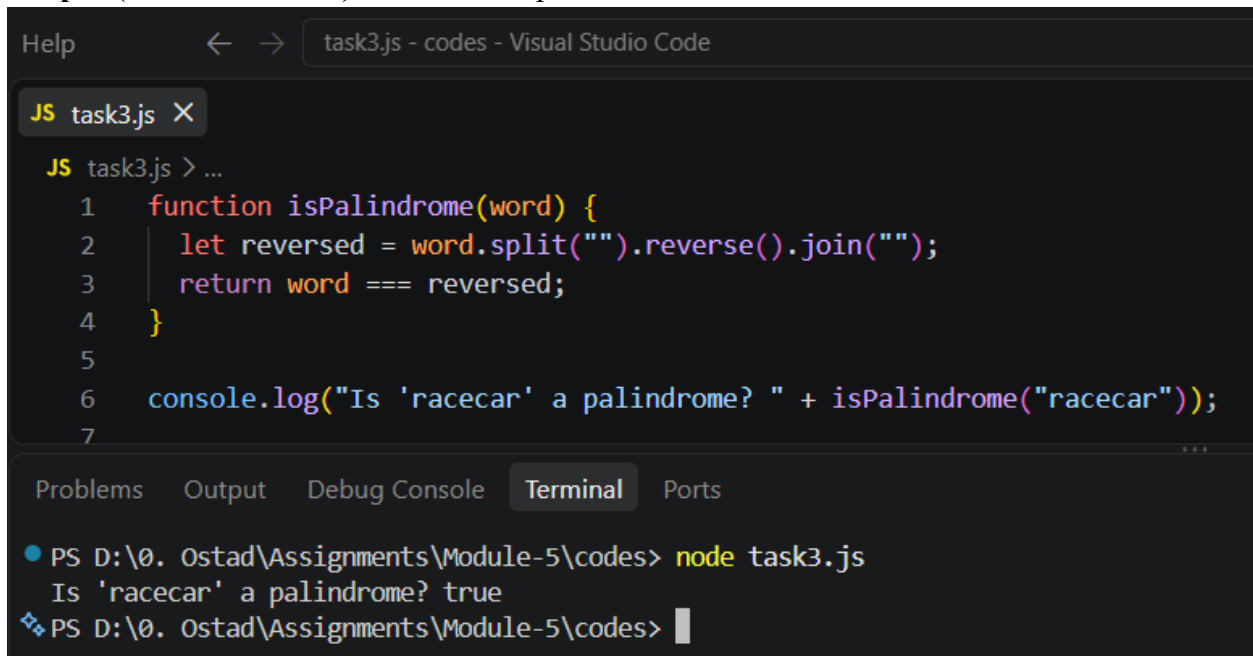
3. Problem Statement:

Check for Palindrome.

Source Code:

```
function isPalindrome(word) {  
  let reversed = word.split("").reverse().join("");  
  return word === reversed;  
}  
  
console.log("Is 'racecar' a palindrome? " + isPalindrome("racecar"));
```

Output (with Screenshot): Is 'racecar' a palindrome? true

The screenshot shows the Visual Studio Code interface. The top bar displays 'task3.js - codes - Visual Studio Code'. The editor window shows the JavaScript code for the palindrome checker. The bottom panel has tabs for 'Problems', 'Output', 'Debug Console', 'Terminal', and 'Ports'. The 'Terminal' tab is active, showing the command 'node task3.js' being executed in a PowerShell prompt, with the output 'Is 'racecar' a palindrome? true'.

```
Help  <  >  task3.js - codes - Visual Studio Code  
  
JS task3.js X  
JS task3.js > ...  
1  function isPalindrome(word) {  
2    let reversed = word.split("").reverse().join("");  
3    return word === reversed;  
4  }  
5  
6  console.log("Is 'racecar' a palindrome? " + isPalindrome("racecar"));  
7  
Problems  Output  Debug Console  Terminal  Ports  
● PS D:\0. Ostad\Assignments\Module-5\codes> node task3.js  
Is 'racecar' a palindrome? true  
PS D:\0. Ostad\Assignments\Module-5\codes> 
```

Explanation:

The word is split into an array of individual letters using `split("")`. That array is then reversed with `reverse()`, and the letters are joined back into a string using `join("")`. If the reversed string matches the original word, it is confirmed to be a palindrome.

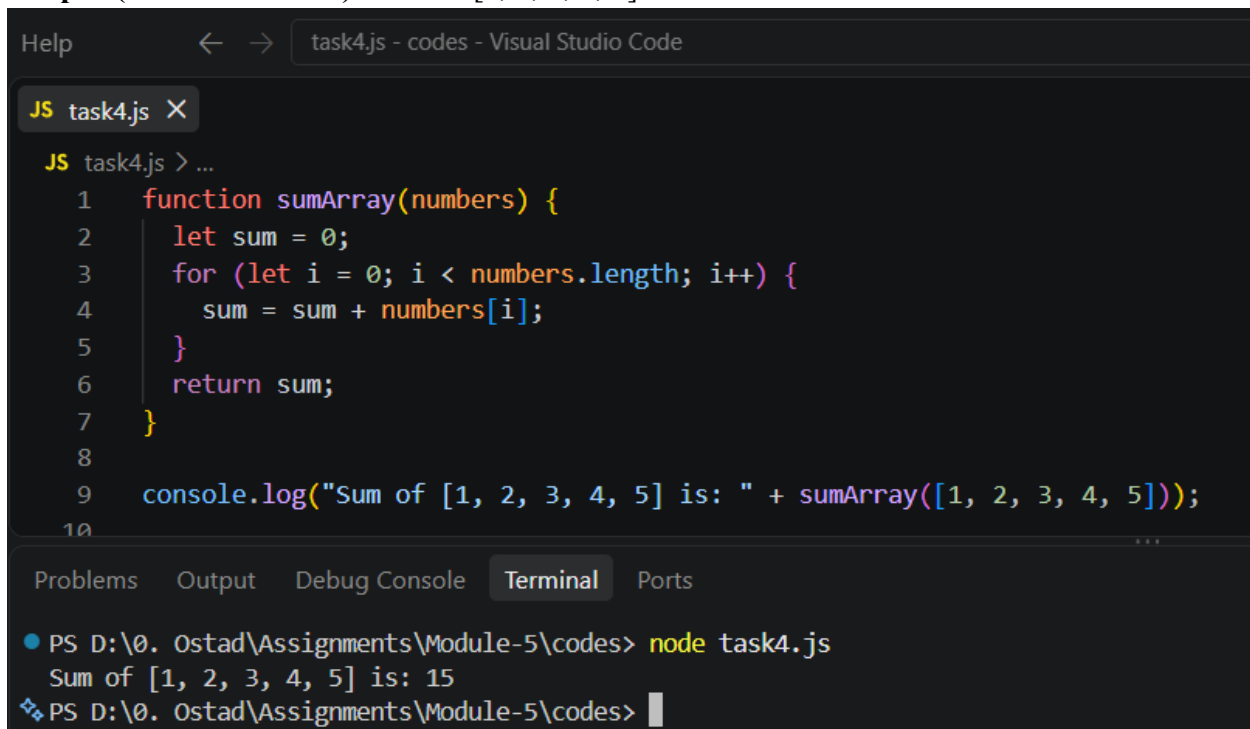
4. Problem Statement:

Sum of Array Elements.

Source Code:

```
function sumArray(numbers) {  
  let sum = 0;  
  for (let i = 0; i < numbers.length; i++) {  
    sum = sum + numbers[i];  
  }  
  return sum;  
}  
  
console.log("Sum of [1, 2, 3, 4, 5] is: " + sumArray([1, 2, 3, 4, 5]));
```

Output (with Screenshot): Sum of [1, 2, 3, 4, 5] is: 15

The screenshot shows the Visual Studio Code editor with a file named 'task4.js'. The code in the editor is a JavaScript function 'sumArray' that takes an array of numbers and returns their sum. Below the code editor, the 'Terminal' tab is active, showing the command 'node task4.js' being executed. The output of the command is 'Sum of [1, 2, 3, 4, 5] is: 15'.

```
Help  ← →  task4.js - codes - Visual Studio Code  
  
JS task4.js X  
JS task4.js > ...  
1  function sumArray(numbers) {  
2    let sum = 0;  
3    for (let i = 0; i < numbers.length; i++) {  
4      sum = sum + numbers[i];  
5    }  
6    return sum;  
7  }  
8  
9  console.log("Sum of [1, 2, 3, 4, 5] is: " + sumArray([1, 2, 3, 4, 5]));  
10  
  
Problems  Output  Debug Console  Terminal  Ports  
● PS D:\0. Ostad\Assignments\Module-5\codes> node task4.js  
Sum of [1, 2, 3, 4, 5] is: 15  
❖ PS D:\0. Ostad\Assignments\Module-5\codes> 
```

Explanation:

The variable sum is initialized to 0. The array is looped through by index, and each element's value is added to sum one at a time. Once the loop is completed, the total is held in sum.

5. Problem Statement:

FizzBuzz (Basic).

Source Code:

```
let output = "";
for (let i = 1; i <= 15; i++) {
  if (i % 3 === 0 && i % 5 === 0) {
    output = output + "FizzBuzz ";
  } else if (i % 3 === 0) {
    output = output + "Fizz ";
  } else if (i % 5 === 0) {
    output = output + "Buzz ";
  } else {
    output = output + i + " ";
  }
}

console.log(output);
```

Output (with Screenshot): 1 2 Fizz 4 Buzz Fizz 7 8 Fizz Buzz 11 Fizz 13 14 FizzBuzz



The screenshot shows a code editor with a file named 'task5.js'. The code is as follows:

```
1 let output = "";
2 for (let i = 1; i <= 15; i++) {
3   if (i % 3 === 0 && i % 5 === 0) {
4     output = output + "FizzBuzz ";
5   } else if (i % 3 === 0) {
6     output = output + "Fizz ";
7   } else if (i % 5 === 0) {
8     output = output + "Buzz ";
9   } else {
10    output = output + i + " ";
11  }
12 }
13
14 console.log(output);
15
```

The terminal output at the bottom shows the command 'node task5.js' being executed, resulting in the output: '1 2 Fizz 4 Buzz Fizz 7 8 Fizz Buzz 11 Fizz 13 14 FizzBuzz'.

Explanation:

Each number from 1 to 15 is checked in the loop. The % (modulo) operator is used to find the remainder $i \% 3 === 0$ means i is divisible by 3 with no remainder left. Divisibility by both 3 and 5 is checked first (otherwise "Fizz" would get printed before the "both" case is ever reached), followed by 3 alone, then 5 alone; otherwise, the number itself is appended. Everything is added to a single output string, so it is printed in one console.log.